CUTLINE INDUSTRIES

AI for software delivery that never sends your code to someone else's model.

Artemis is a mixture-of-experts transformer we train ourselves and run entirely inside the customer's own infrastructure.

01 — PROBLEM

CI failure triage is high-volume, low-complexity toil.

> Every engineering team runs continuous integration, and a large share of what it reports is mechanical to diagnose: a dependency drifted, a flaky test re-ran, a migration ordered wrong, a cache key went stale.
>
> Diagnosing these is not hard. It is **tedious**, it interrupts whoever is on call, and it scales linearly with the number of pipelines a team runs.

02 — WHY THAT MARKET IS BLOCKED

The tools that automate it require source-code egress.

Existing assistants send source code and build logs to a third-party model API. For a large class of buyer, that single architectural fact disqualifies the entire category:

Regulated industries

Finance, healthcare, and critical infrastructure operate under audit regimes that treat third-party model inference as uncontrolled data processing.

Defence & public sector

Suppliers working under export control or classification cannot route source artefacts through a commercial inference endpoint.

Source-code escrow

Contractual custody obligations are incompatible with transmitting code to a provider the customer has no privity with.

A self-hosted engine for the software delivery lifecycle.

01

CI failure triage

Reads the build log, the diff, and the test output; identifies the failing stage and the probable cause.

02

Deploy review

Inspects a release against prior deploy outcomes and flags the changes that historically preceded rollbacks.

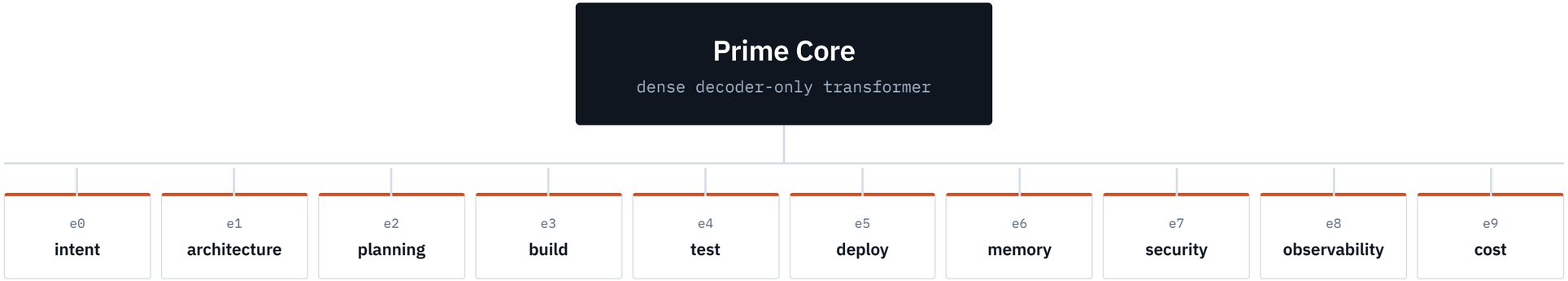
03

Cost analysis

Attributes build and infrastructure spend to the pipelines and services that generate it.

The deployment model is the product. Artemis runs on the customer's own hardware. No source code, build log, or telemetry leaves their environment, and no external model provider is called at inference time.

Prime Core, plus ten additively-routed experts.



$$h \leftarrow x + \mathrm{ffn}(h) + \Sigma\ g_e \cdot \mathrm{expert}_e(h)$$

Experts are **additive**, not a replacement for the dense feed-forward path — the core always contributes. Each expert maps to one stage of delivery, so routing is supervisable against the stage a task actually belongs to.

05 — VALIDATION

What is built, trained, and measured today.

COMPONENT	STATUS	EVIDENCE
Prime Core + 10 MoE experts	**Trained**	153M-parameter checkpoint, trained to its compute-optimal (Chinchilla) token budget
Architecture justification	**Measured**	Matched-compute dense baseline and a routing ablation, both run and evaluated
Corpus pipeline	**Production**	Resumable, bounded-disk tokeniser with byte-offset rollback and SHA-256 dedup; 2.35× on 4 cores
Inference gateway	**Deployed**	Streaming API in production; credentialed CORS and CSRF double-submit, verified live
Sharded training (FSDP)	**Written**	Implemented against PyTorch `fully_shard`; validated on CPU/gloo, **not yet on GPUs**

The model is deliberately small. The purpose of this checkpoint was to establish that the routed architecture earns its parameters against a matched-compute dense baseline before scaling it — not to demonstrate capability.

Every run to date has been on a free Tesla T4.

- **T4 is Turing (SM 7.5).** No native bfloat16 — `is_bf16_supported()` returns true, but the path is emulated, so training runs in fp16 with loss scaling.
- **16 GB of VRAM** caps us near 150M parameters. Our last attempt at a larger batch returned CUDA OOM at 14.56 GiB.
- **Single device.** The FSDP sharding path cannot be validated at all, because there is no second GPU to shard across.

WHAT AMPERE-OR-NEWER UNLOCKS

bf16 tensor cores (SM ≥ 8.0) — numerically stable training without loss scaling.

Multi-GPU interconnect — validates the sharded path that a 1–3B model requires.

TensorRT / Triton — quantised serving, which is a direct margin driver for a self-hosted product.

A100-class access is the difference between a 153M research artefact and a 1–3B production model. It is the single gating dependency on this project.

07 — MARKET

DevOps tooling, entered through the segment that cannot buy the incumbents.

WEDGE

No-egress software delivery

Teams for whom third-party model inference is disqualifying, not merely undesirable. They are underserved precisely because the constraint is architectural and cannot be retrofitted by a competitor built on a hosted API.

WHY A SMALL MODEL IS THE RIGHT ANSWER HERE

Narrow task, structured context

CI triage has bounded inputs — a log, a diff, a test report. Cost per triaged failure, not general capability, is the metric buyers compare. A specialised 1–3B model competes on that metric in a way a frontier model's pricing does not.

We are pre-revenue and make no forecast here. The claim on this slide is about why the segment exists and why it is structurally hard for an API-based competitor to serve — not about captured demand.

08 — ASK

GPU access. Not capital.

A100

Ampere-class compute

To train the 1–3B production model to its compute-optimal budget, and to validate the FSDP path on real multi-GPU hardware before it costs money to debug.

SDK

TensorRT & Triton

Quantised inference serving. Because Artemis is self-hosted, inference efficiency is a margin driver rather than an infrastructure line item.

DGX

Technical guidance

Review of the sharded training configuration. We have the implementation; we have never been able to run it on the hardware it targets.

> We are not raising. The binding constraint on this project is not capital — it is that we have never had access to a GPU with native bf16 and a second device to shard across.

09 — COMPANY

Cutline Industries

Entity	**Registered LLC**
Founded	**Under 10 years**
Team	**1 — founder-operated**
Funding	**Bootstrapped, pre-seed**
Revenue	**Pre-revenue**
Prior cloud credits	**None**

- ✓ Not a consultancy or outsourced development firm
- ✓ Not crypto-associated
- ✓ Not a cloud service provider, reseller, or distributor
- ✓ Not a public company
- ✓ Working website, actively operated by a founder

> Artemis exists because one architectural decision — never call an external model — was treated as a product requirement rather than a limitation. Everything else followed from it.

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